Active Forgetting via Dedicated Molecular Signaling Pathways (Davis & Zhong 2017)

Review of neurobiological evidence showing that forgetting is not passive decay but an active cellular process governed by specific signaling pathways (including Rac1 and dopamine signaling) required for behavioral flexibility.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL, PHILOSOPHICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Ronald Davis and Yi Zhong (2017) synthesized groundbreaking molecular, genetic, and optogenetic evidence demonstrating that biological forgetting is not passive decay or simple failure to access stored engrams. Instead, cells possess dedicated, metabolically active biochemical pathways whose primary function is to dismantle and suppress synaptic memory traces. In Drosophila and mammalian models, activation of the small G protein Rac1 and specific dopamine receptors actively promotes memory decay. When these active forgetting cascades are genetically or pharmacologically inhibited, animals retain memories for unnaturally extended periods but display severe impairments in behavioral flexibility, reversal learning, and cognitive updating when environmental contingencies shift.

EVIDENCE AND CAUSAL ASSESSMENT

Davis and Zhong categorized active forgetting into two major forms:

1. Intrinsic Forgetting: Ongoing, constitutive cellular processes where default molecular activity (such as basal Rac1/Cdc42 signaling) steadily erodes synaptic potentiations unless counteracted by memory consolidation loops.

2. Neurogenesis-Mediated and Induced Forgetting: Environmentally or signal-triggered cascades (such as bursts of dopamine during unexpected events or adult hippocampal neurogenesis) that accelerate the dismantling of specific engrams.

3. Causal Validation: Optogenetic stimulation of dopamine neurons or biochemical activation of Rac1 accelerates forgetting of conditioned odor avoidance in fruit flies; conversely, dominant-negative Rac1 inhibition prolongs memory retention while causing catastrophic failure when flies are trained on reversed reward associations.

EFFECT SIZE AND PERSISTENCE

In Drosophila assays, activating the Rac1 signaling pathway halves memory half-life, whereas inhibiting Rac1 extends memory retention from hours to days. In rodent Morris water maze reversal tasks, animals with intact active forgetting adapt to relocated platforms within 2 trials, whereas animals with disrupted forgetting pathways perseverate on original coordinates across 8 or more trials.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as robust (<20%) in controlled laboratory model organisms. While human memory involves vast autobiographical and cultural layers, identical cellular signaling families (Rac1, actin-remodeling GTPases, dopamine D1/D2 receptor dynamics) are conserved in human neocortex and hippocampus, suggesting an ancient evolutionary imperative for active erasure.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Active forgetting does not imply that human forgetting is always intentional, controllable, or adaptive. Clinical neurodegenerative pathologies (such as tauopathies or amyloid plaque accumulation) represent catastrophic structural damage rather than regulated molecular transience.

WHAT ACTION IS REASONABLY JUSTIFIED

Artificial cognitive architectures and lifelong learning systems should incorporate active, signal-driven weight erasure or regularization pathways (analogous to Rac1 activation) rather than relying exclusively on passive weight decay or static context caps.

VERIFIED SOURCES & DOI IDENTIFIERS

- Davis, R. L., & Zhong, Y. (2017). The Biology of Forgetting—A Perspective. Neuron, 95(3), 490–503. https://doi.org/10.1016/j.neuron.2017.05.039